

Supplementary Materials

Computational evidence for a distributed AT-based regulatory mechanism enabling de novo gene birth in *Pandoravirus salinus*

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Supplementary Table S1

Table S1. Codon usage comparison between annotated and ORFan genes: RSCU, GC3, GC12, ENC, and CAI with statistical tests.

Metric	Annotate d_n	Annotate d_mean	Annotate d_sd	ORFan_n	ORFan_ mean	ORFan_s d	Differenc e
GC content	528	0.6502	0.0321	902	0.6428	0.0427	+0.0074
GC3 (wobble position)	528	0.7497	0.0574	902	0.7343	0.0774	+0.0155
GC12 (constrai ned positions)	528	0.6021	0.0337	902	0.5998	0.0428	+0.0023
ENC (effective number of codons)	528	44.9559	4.9847	902	45.9831	6.3753	-1.0272
CAI (codon adaptati on index)	528	0.7913	0.0273	902	0.7837	0.0337	+0.0075

Supplementary Table S2

Table S2. ESM-2 structural family assignments for 170 representative proteins, including tier, compositional cluster, and nearest-neighbor similarity.

Gene_ID	Structural_family	Confidence	Product	GC_content	Gene_length_bp	Start_pos	End_pos	Strand	Notes
1082	structural_cluster_1	0.524	hypothetical protein	0.5236	552	34572	35123	-	tier=A, cluster_size=53
1122	structural_cluster_1	0.546	hypothetical protein	0.5465	333	102904	103236	+	tier=A, cluster_size=53
1131	structural_cluster_1	0.551	hypothetical protein	0.5511	372	113160	113531	-	tier=A, cluster_size=53
1140	structural_cluster_1	0.539	hypothetical protein	0.5386	531	131036	131566	+	tier=A, cluster_size=53
1188	structural_cluster_1	0.532	hypothetical protein	0.5318	393	209222	209614	-	tier=A, cluster_size=53
1254	structural_cluster_1	0.545	hypothetical protein	0.5450	1044	320553	321596	+	tier=A, cluster_size=53
1325	structural_cluster_1	0.536	hypothetical protein	0.5360	375	464920	465294	-	tier=A, cluster_size=53
1328	structural_cluster_1	0.525	hypothetical protein	0.5246	852	468298	469149	+	tier=A, cluster_size=53
1329	structural_cluster_1	0.558	hypothetical protein	0.5585	684	469447	470130	+	tier=A, cluster_size=53
1338	structural_cluster_1	0.564	hypothetical	0.5638	447	494215	494661	+	tier=A, cluster

Gene_ID	Structural_family	Confidence	Product	GC_content	Gene_length_bp	Start_pos	End_pos	Strand	Notes
	uster_1		protein						_size=53
1356	structural_cluster_1	0.521	hypothetical protein	0.5206	315	527479	527793	-	tier=A, cluster_size=53
1394	structural_cluster_1	0.522	hypothetical protein	0.5220	387	594682	595068	-	tier=A, cluster_size=53
1407	structural_cluster_1	0.542	hypothetical protein	0.5425	153	614585	614737	-	tier=A, cluster_size=53
1448	structural_cluster_1	0.524	hypothetical protein	0.5245	387	694291	694677	+	tier=A, cluster_size=53
1478	structural_cluster_1	0.559	hypothetical protein	0.5590	288	748176	748463	+	tier=A, cluster_size=53
1520	structural_cluster_1	0.497	hypothetical protein	0.4970	330	823500	823829	-	tier=A, cluster_size=53
1551	structural_cluster_1	0.556	hypothetical protein	0.5556	243	876417	876659	-	tier=A, cluster_size=53
1635	structural_cluster_1	0.550	hypothetical protein	0.5505	327	1018138	1018464	+	tier=A, cluster_size=53
1726	structural_cluster_1	0.534	hypothetical protein	0.5342	234	1160705	1160938	+	tier=A, cluster_size=53

Gene_ID	Structural_family	Confidence	Product	GC_content	Gene_length_bp	Start_pos	End_pos	Strand	Notes
1773	structural_cluster_1	0.551	hypothetical protein	0.5514	321	1253057	1253377	-	tier=A, cluster_size=53
1800	structural_cluster_1	0.562	hypothetical protein	0.5616	276	1304665	1304940	+	tier=A, cluster_size=53
2002	structural_cluster_1	0.564	hypothetical protein	0.5637	408	1655327	1655734	+	tier=A, cluster_size=53
2278	structural_cluster_1	0.543	hypothetical protein	0.5433	543	2125529	2126071	+	tier=A, cluster_size=53
2281	structural_cluster_1	0.541	hypothetical protein	0.5407	246	2130320	2130565	+	tier=A, cluster_size=53
2334	structural_cluster_1	0.557	hypothetical protein	0.5570	237	2216131	2216367	+	tier=A, cluster_size=53
2341	structural_cluster_1	0.592	hypothetical protein	0.5915	306	2229638	2229943	+	tier=C, cluster_size=53
2362	structural_cluster_1	0.548	hypothetical protein	0.5478	345	2266329	2266673	+	tier=A, cluster_size=53
2365	structural_cluster_1	0.530	hypothetical protein	0.5302	315	2270179	2270493	+	tier=A, cluster_size=53
2373	structural_cluster_1	0.524	hypothetical	0.5245	204	2281789	2281992	-	tier=A, cluster

Gene_ID	Structural_family	Confidence	Product	GC_content	Gene_length_bp	Start_pos	End_pos	Strand	Notes
	uster_1		protein						_size=53
2401	structural_cluster_1	0.604	hypothetical protein	0.6044	225	2325113	2325337	+	tier=C, cluster_size=53
2425	structural_cluster_1	0.546	hypothetical protein	0.5464	366	2363321	2363686	+	tier=A, cluster_size=53
2465	structural_cluster_1	0.533	hypothetical protein	0.5330	651	2436910	2437560	-	tier=A, cluster_size=53
2470	structural_cluster_1	0.565	hypothetical protein	0.5651	384	2442902	2443285	+	tier=A, cluster_size=53
2473	structural_cluster_1	0.545	hypothetical protein	0.5446	336	2445291	2445626	+	tier=A, cluster_size=53
2475	structural_cluster_1	0.519	hypothetical protein	0.5190	237	2447673	2447909	-	tier=A, cluster_size=53
2477	structural_cluster_1	0.488	hypothetical protein	0.4883	213	2449201	2449413	-	tier=A, cluster_size=53
2478	structural_cluster_1	0.543	hypothetical protein	0.5427	960	2449497	2450456	-	tier=A, cluster_size=53
2479	structural_cluster_1	0.544	hypothetical protein	0.5440	579	2451085	2451663	-	tier=A, cluster_size=53

Gene_ID	Structural_family	Confidence	Product	GC_content	Gene_length_bp	Start_pos	End_pos	Strand	Notes
2481	structural_cluster_1	0.536	hypothetical protein	0.5365	192	2453815	2454006	-	tier=A, cluster_size=53
2483	structural_cluster_1	0.514	hypothetical protein	0.5142	387	2455673	2456059	-	tier=A, cluster_size=53
2492	structural_cluster_1	0.567	hypothetical protein	0.5669	381	2469070	2469450	+	tier=A, cluster_size=53
1173	structural_cluster_10	0.606	hypothetical protein	0.6056	715	186765	187479	-	tier=C, cluster_size=5
1516	structural_cluster_10	0.566	hypothetical protein	0.5657	198	817589	817786	-	tier=A, cluster_size=5
1945	structural_cluster_10	0.678	hypothetical protein	0.6782	348	1545443	1545790	+	tier=C, cluster_size=5
2121	structural_cluster_10	0.581	hypothetical protein	0.5809	303	1857627	1857929	+	tier=C, cluster_size=5
2487	structural_cluster_10	0.458	hypothetical protein	0.4583	168	2459134	2459301	+	tier=A, cluster_size=5
1186	structural_cluster_11	0.673	hypothetical protein	0.6728	651	206557	207207	+	tier=C, cluster_size=10
1211	structural_cluster_11	0.727	hypothetical	0.7268	981	245592	246572	+	tier=C, cluster_size=10

Gene_ID	Structural_family	Confidence	Product	GC_content	Gene_length_bp	Start_pos	End_pos	Strand	Notes
	uster_11		protein						_size=10
1284	structural_cluster_11	0.693	hypothetical protein	0.6933	1653	380606	382258	-	tier=C, cluster_size=10
1314	structural_cluster_11	0.673	hypothetical protein	0.6734	1044	441428	442471	+	tier=C, cluster_size=10
1317	structural_cluster_11	0.630	hypothetical protein	0.6300	600	447603	448202	-	tier=C, cluster_size=10
1384	structural_cluster_11	0.682	hypothetical protein	0.6818	2445	578439	580883	+	tier=C, cluster_size=10
1415	structural_cluster_11	0.678	hypothetical protein	0.6778	717	628822	629538	-	tier=C, cluster_size=10
1477	structural_cluster_11	0.728	hypothetical protein	0.7278	900	746560	747459	-	tier=C, cluster_size=10
1482	structural_cluster_11	0.703	hypothetical protein	0.7030	2172	753904	756075	+	tier=C, cluster_size=10
1570	structural_cluster_11	0.682	hypothetical protein	0.6823	702	910229	910930	+	tier=C, cluster_size=10
1081	structural_cluster_12	0.614	hypothetical protein	0.6140	987	33485	34471	-	tier=C, cluster_size=24

Gene_ID	Structural_family	Confidence	Product	GC_content	Gene_length_bp	Start_pos	End_pos	Strand	Notes
1129	structural_cluster_12	0.548	hypothetical protein	0.5482	228	111192	111419	+	tier=A, cluster_size=24
1171	structural_cluster_12	0.671	hypothetical protein	0.6706	1524	182489	184012	+	tier=C, cluster_size=24
1200	structural_cluster_12	0.704	hypothetical protein	0.7037	486	226134	226619	-	tier=C, cluster_size=24
1210	structural_cluster_12	0.688	hypothetical protein	0.6884	1335	243754	245088	+	tier=C, cluster_size=24
1216	structural_cluster_12	0.608	hypothetical protein	0.6077	915	256613	257527	-	tier=C, cluster_size=24
1220	structural_cluster_12	0.645	hypothetical protein	0.6452	699	263923	264621	+	tier=C, cluster_size=24
1367	structural_cluster_12	0.691	hypothetical protein	0.6912	651	545339	545989	+	tier=C, cluster_size=24
1408	structural_cluster_12	0.700	hypothetical protein	0.7002	3462	615056	618517	-	tier=C, cluster_size=24
1463	structural_cluster_12	0.701	hypothetical protein	0.7014	1209	722030	723238	+	tier=C, cluster_size=24
1507	structural_cluster_12	0.672	hypothetical	0.6723	1584	803347	804930	+	tier=C, cluster

Gene_ID	Structural_family	Confidence	Product	GC_content	Gene_length_bp	Start_pos	End_pos	Strand	Notes
	uster_12		protein						_size=24
1544	structural_cluster_12	0.731	hypothetical protein	0.7313	3048	863336	866383	-	tier=B, cluster_size=24
1675	structural_cluster_12	0.650	hypothetical protein	0.6501	1389	1076826	1078214	-	tier=C, cluster_size=24
1737	structural_cluster_12	0.718	hypothetical protein	0.7179	507	1184184	1184690	+	tier=C, cluster_size=24
1793	structural_cluster_12	0.621	hypothetical protein	0.6206	1497	1293151	1294647	+	tier=C, cluster_size=24
1891	structural_cluster_12	0.612	hypothetical protein	0.6118	1149	1447224	1448372	-	tier=C, cluster_size=24
1949	structural_cluster_12	0.701	hypothetical protein	0.7010	612	1557848	1558459	+	tier=C, cluster_size=24
2095	structural_cluster_12	0.591	hypothetical protein	0.5913	504	1815330	1815833	-	tier=C, cluster_size=24
2145	structural_cluster_12	0.677	hypothetical protein	0.6771	1344	1896578	1897921	+	tier=C, cluster_size=24
2244	structural_cluster_12	0.626	hypothetical protein	0.6262	915	2065533	2066447	+	tier=C, cluster_size=24

Gene_ID	Structural_family	Confidence	Product	GC_content	Gene_length_bp	Start_pos	End_pos	Strand	Notes
2251	structural_cluster_12	0.682	hypothetical protein	0.6815	1482	2080899	2082380	+	tier=B, cluster_size=24
2328	structural_cluster_12	0.692	hypothetical protein	0.6917	918	2205779	2206696	-	tier=C, cluster_size=24
2403	structural_cluster_12	0.623	hypothetical protein	0.6227	1092	2326612	2327703	-	tier=C, cluster_size=24
2484	structural_cluster_12	0.686	hypothetical protein	0.6856	264	2456478	2456741	-	tier=C, cluster_size=24
1107	structural_cluster_3	0.639	hypothetical protein	0.6386	1392	74035	75426	+	tier=C, cluster_size=2
1113	structural_cluster_5	0.651	hypothetical protein	0.6507	942	85168	86109	+	tier=C, cluster_size=9
1454	structural_cluster_5	0.510	hypothetical protein	0.5104	288	708218	708505	+	tier=A, cluster_size=9
1642	structural_cluster_5	0.588	hypothetical protein	0.5878	621	1026727	1027347	-	tier=C, cluster_size=9
1647	structural_cluster_5	0.618	hypothetical protein	0.6185	540	1034087	1034626	+	tier=C, cluster_size=9
1876	structural_cluster	0.545	hypothetical	0.5450	411	1419235	1419645	-	tier=A, cluster

Gene_ID	Structural_family	Confidence	Product	GC_content	Gene_length_bp	Start_pos	End_pos	Strand	Notes
	uster_5		protein						_size=9
2264	structural_cluster_5	0.595	hypothetical protein	0.5952	294	2101492	2101785	+	tier=C, cluster_size=9
2322	structural_cluster_5	0.577	hypothetical protein	0.5769	234	2196339	2196572	+	tier=C, cluster_size=9
2400	structural_cluster_5	0.604	hypothetical protein	0.6042	240	2323872	2324111	+	tier=C, cluster_size=9
1238	structural_cluster_6	0.576	hypothetical protein	0.5761	276	292761	293036	+	tier=C, cluster_size=4
2074	structural_cluster_6	0.561	hypothetical protein	0.5614	285	1777968	1778252	+	tier=A, cluster_size=4
2258	structural_cluster_6	0.540	hypothetical protein	0.5402	709	2092298	2093006	-	tier=A, cluster_size=4
2457	structural_cluster_6	0.644	hypothetical protein	0.6440	663	2421860	2422522	+	tier=C, cluster_size=4
1133	structural_cluster_8	0.662	hypothetical protein	0.6616	1188	116789	117976	-	tier=C, cluster_size=38
1157	structural_cluster_8	0.690	hypothetical protein	0.6901	726	157805	158530	-	tier=B, cluster_size=38

Gene_ID	Structural_family	Confidence	Product	GC_content	Gene_length_bp	Start_pos	End_pos	Strand	Notes
1308	structural_cluster_8	0.645	hypothetical protein	0.6448	2787	423898	426684	-	tier=C, cluster_size=38
1459	structural_cluster_8	0.652	hypothetical protein	0.6515	264	714788	715051	+	tier=C, cluster_size=38
1460	structural_cluster_8	0.740	Nucleoporin FG2 superfamily incomplete domain	0.7404	2088	715840	717927	-	tier=B, cluster_size=38
1501	structural_cluster_8	0.692	hypothetical protein	0.6925	696	792579	793274	-	tier=C, cluster_size=38
1585	structural_cluster_8	0.654	hypothetical protein	0.6541	1668	938544	940211	+	tier=C, cluster_size=38
1615	structural_cluster_8	0.622	hypothetical protein	0.6223	1149	988696	989844	-	tier=C, cluster_size=38
1632	structural_cluster_8	0.650	ankyrin repeat domain	0.6501	1266	1013066	1014331	-	tier=B, cluster_size=38
1661	structural_cluster_8	0.627	hypothetical protein	0.6267	1176	105719	1058294	+	tier=C, cluster_size=38

Gene_ID	Structural_family	Confidence	Product	GC_content	Gene_length_bp	Start_pos	End_pos	Strand	Notes
1770	structural_cluster_8	0.679	serine-threonine kinase	0.6786	5439	1244600	1250038	+	tier=B, cluster_size=38
1779	structural_cluster_8	0.650	Dihydrofolate reductase incomplete domain	0.6504	3567	1259681	1263247	-	tier=B, cluster_size=38
1781	structural_cluster_8	0.669	hypothetical protein	0.6689	912	1269247	1270158	-	tier=C, cluster_size=38
1810	structural_cluster_8	0.636	hypothetical protein	0.6355	609	1321715	1322323	-	tier=C, cluster_size=38
1885	structural_cluster_8	0.560	Ubiquitin	0.5605	2917	1431566	1434482	+	tier=A, cluster_size=38
1958	structural_cluster_8	0.661	hypothetical protein	0.6611	1676	1570847	1572522	+	tier=C, cluster_size=38
1966	structural_cluster_8	0.663	ankyrin repeat domain	0.6629	1587	1586661	1588247	-	tier=B, cluster_size=38
1971	structural_cluster_8	0.641	hypothetical protein	0.6412	2238	1595992	1598229	-	tier=C, cluster_size=38
1997	struct	0.635	hypoth	0.6352	603	16482	16488	+	tier=C,

Gene_ID	Structural_family	Confidence	Product	GC_content	Gene_length_bp	Start_pos	End_pos	Strand	Notes
	ural_cluster_8		etical protein			71	73		cluster_size=38
2028	structural_cluster_8	0.630	ankyrin repeat domain	0.6295	2232	1702376	1704607	+	tier=B, cluster_size=38
2033	structural_cluster_8	0.626	hypothetical protein	0.6260	861	1712679	1713539	-	tier=C, cluster_size=38
2034	structural_cluster_8	0.620	ankyrin repeat domain	0.6204	1233	1713922	1715154	-	tier=B, cluster_size=38
2043	structural_cluster_8	0.559	UBC superfamily incomplete domain	0.5587	528	1730907	1731434	+	tier=A, cluster_size=38
2060	structural_cluster_8	0.615	hypothetical protein	0.6152	1302	1757520	1758821	-	tier=C, cluster_size=38
2064	structural_cluster_8	0.626	hypothetical protein	0.6261	1380	1764364	1765743	+	tier=C, cluster_size=38
2099	structural_cluster_8	0.555	hypothetical protein	0.5546	1291	1822122	1823412	+	tier=A, cluster_size=38
2135	structural_cluster_8	0.584	hypothetical	0.5841	678	1879410	1880087	-	tier=C, cluster_size=

Gene_ID	Structural_family	Confidence	Product	GC_content	Gene_length_bp	Start_pos	End_pos	Strand	Notes
	8		protein						38
2149	structural_cluster_8	0.659	hypothetical protein	0.6588	1911	1904717	1906627	-	tier=C, cluster_size=38
2156	structural_cluster_8	0.657	ankyrin repeat domain	0.6573	1824	1916831	1918654	+	tier=B, cluster_size=38
2170	structural_cluster_8	0.692	Fascin-like incomplete domain	0.6920	1422	1940682	1942103	+	tier=B, cluster_size=38
2198	structural_cluster_8	0.700	hypothetical protein	0.6999	723	1994604	1995326	+	tier=C, cluster_size=38
2220	structural_cluster_8	0.650	hypothetical protein	0.6499	1948	2031028	2032975	-	tier=C, cluster_size=38
2223	structural_cluster_8	0.552	S-phase kinase - associated protein 1	0.5517	687	2035314	2036000	+	tier=A, cluster_size=38
2238	structural_cluster_8	0.662	hypothetical protein	0.6617	999	2054071	2055069	-	tier=C, cluster_size=38
2249	structural_cluster_8	0.650	hypothetical	0.6505	1173	2077105	2078277	-	tier=C, cluster_size=

Gene_ID	Structural_family	Confidence	Product	GC_content	Gene_length_bp	Start_pos	End_pos	Strand	Notes
	8		protein						38
2295	structural_cluster_8	0.689	hypothetical protein	0.6889	495	2151287	2151781	+	tier=C, cluster_size=38
2296	structural_cluster_8	0.679	hypothetical protein	0.6786	585	2152084	2152668	+	tier=C, cluster_size=38
1105	structural_cluster_9	0.716	hypothetical protein	0.7165	642	70710	71351	-	tier=C, cluster_size=22
1110	structural_cluster_9	0.686	hypothetical protein	0.6859	3438	78087	81524	-	tier=C, cluster_size=22
1185	structural_cluster_9	0.721	hypothetical protein	0.7207	999	205227	206225	+	tier=C, cluster_size=22
1222	structural_cluster_9	0.670	hypothetical protein	0.6703	831	266577	267407	-	tier=C, cluster_size=22
1258	structural_cluster_9	0.653	hypothetical protein	0.6532	2886	328292	331177	-	tier=C, cluster_size=22
1374	structural_cluster_9	0.665	hypothetical protein	0.6646	1419	559644	561062	+	tier=B, cluster_size=22
1391	structural_cluster_9	0.653	hypothetical protein	0.6527	1365	588641	590005	-	tier=C, cluster_size=22
1398	struct	0.652	hypoth	0.6521	1995	59927	60127	-	tier=C,

Gene_ID	Structural_family	Confidence	Product	GC_content	Gene_length_bp	Start_pos	End_pos	Strand	Notes
	ural_cluster_9		etical protein			8	2		cluster_size=22
1402	structural_cluster_9	0.662	hypothetical protein	0.6625	951	605950	606900	-	tier=C, cluster_size=22
1432	structural_cluster_9	0.696	hypothetical protein	0.6965	1140	667381	668520	-	tier=C, cluster_size=22
1575	structural_cluster_9	0.716	hypothetical protein	0.7158	2730	916460	919189	-	tier=C, cluster_size=22
1678	structural_cluster_9	0.615	hypothetical protein	0.6147	1308	1081728	1083035	+	tier=C, cluster_size=22
1740	structural_cluster_9	0.675	hypothetical protein	0.6746	1755	1187084	1188838	-	tier=C, cluster_size=22
2093	structural_cluster_9	0.592	hypothetical protein	0.5917	1129	1811155	1812283	+	tier=C, cluster_size=22
2103	structural_cluster_9	0.653	hypothetical protein	0.6531	1401	1828440	1829840	-	tier=C, cluster_size=22
2163	structural_cluster_9	0.642	hypothetical protein	0.6419	849	1928973	1929821	-	tier=C, cluster_size=22
2168	structural_cluster_9	0.634	hypothetical protein	0.6338	2127	1936573	1938699	+	tier=C, cluster_size=

Gene_ID	Structural_family	Confidence	Product	GC_content	Gene_length_bp	Start_pos	End_pos	Strand	Notes
	9								22
2206	structural_cluster_9	0.640	hypothetical protein	0.6400	1089	200711	2008199	+	tier=C, cluster_size=22
2282	structural_cluster_9	0.659	hypothetical protein	0.6593	1356	2131572	2132927	+	tier=C, cluster_size=22
2291	structural_cluster_9	0.636	hypothetical protein	0.6364	1837	2145858	2147694	-	tier=C, cluster_size=22
2435	structural_cluster_9	0.682	hypothetical protein	0.6822	2130	2380974	2383103	-	tier=C, cluster_size=22
2444	structural_cluster_9	0.696	hypothetical protein	0.6958	1542	2397464	2399005	+	tier=C, cluster_size=22
1215	structural_singleton	0.608	hypothetical protein	0.6080	1449	254476	255924	+	tier=B, cluster_size=1
2185	structural_singleton	0.653	hypothetical protein	0.6527	429	1970243	1970671	-	tier=C, cluster_size=1
2257	structural_singleton	0.522	hypothetical protein	0.5217	646	2091681	2092326	+	tier=A, cluster_size=1

Supplementary Table S3

Table S3. Gene family detection results: all 14 compositional families with member counts, mean similarity, and spatial distribution.

Gene_ID	Family	Similarity_score	Product	GC_content	Gene_length_bp	Genomic_position_bp
1215	family_1040	0.648	hypothetical protein	0.6080	1449	254476
2356	family_1040	0.648	hypothetical protein	0.6934	897	2255397
2250	family_1197	0.503	hypothetical protein	0.6710	1602	2078759
2251	family_1197	0.503	hypothetical protein	0.6815	1482	2080899
170	family_13	0.910	tail fiber protein	0.4140	2838	86458
302	family_13	0.910	collagen-like protein 2	0.3764	4788	237025
345	family_13	0.910	collagen-like protein 3	0.3926	2820	292697
347	family_13	0.910	tail fiber protein	0.4051	2439	298179
803	family_13	0.910	collagen-like protein 6	0.3357	4164	875974
804	family_13	0.910	collagen-like protein 7	0.3758	5814	880269
1779	family_1324	0.544	Dihydrofolate reductase incomplete domain	0.6504	3567	1259681
1807	family_1324	0.544	Dihydrofolate reductase incomplete domain	0.6399	3513	1314413
1770	family_1325	0.532	serine-threonine kinase	0.6786	5439	1244600

Gene_ID	Family	Similarity_score	Product	GC_content	Gene_length_bp	Genomic_position_bp
1948	family_1325	0.532	serine-threonine kinase	0.6941	6483	1549807
1450	family_1339	0.502	hypothetical protein	0.7202	4392	696982
1544	family_1339	0.502	hypothetical protein	0.7313	3048	863336
77	family_18	0.991	holin/anti-holin	0.4815	324	45186
78	family_18	0.991	holin/anti-holin	0.4843	318	45192
182	family_252	0.890	transposase	0.3229	573	100219
916	family_252	0.890	transposase	0.3106	573	1009063
100	family_3	0.910	hypothetical protein	0.2963	135	22736
1062	family_3	0.910	hypothetical protein	0.2833	180	1180934
1460	family_402	0.553	Nucleoporin FG2 superfamily incomplete domain	0.7404	2088	715840
1496	family_402	0.553	hypothetical protein	0.6977	2286	783261
1157	family_407	0.630	hypothetical protein	0.6901	726	157805
1776	family_407	0.630	CBM6-CBM35-CBM36 like incomplete domain	0.6708	729	1256688
1204	family_419	0.575	Fascin-like incomplete	0.6810	1398	233934

Gene_ID	Family	Similarity_score	Product	GC_content	Gene_length_bp	Genomic_position_bp
1561	family_419	0.575	Fascin-like incomplete domain	0.6880	1737	895727
1704	family_419	0.575	Fascin-like incomplete domain	0.6751	1416	1126238
1938	family_419	0.575	Fascin-like incomplete domain	0.6738	1398	1529707
1943	family_419	0.575	Fascin-like protein	0.7104	1236	1540514
1972	family_419	0.575	Fascin-like incomplete domain	0.6830	1407	1598636
1973	family_419	0.575	Fascin-like protein	0.7034	1335	1600745
1975	family_419	0.575	Fascin-like incomplete domain	0.7036	1407	1605142
1976	family_419	0.575	Fascin-like incomplete domain	0.6931	1401	1607077
2154	family_419	0.575	Fascin-like incomplete domain	0.6910	1395	1913822
2170	family_419	0.575	Fascin-like incomplete domain	0.6920	1422	1940682
2171	family_419	0.575	Fascin-like incomplete domain	0.6884	1383	1942644
2028	family_706	0.579	ankyrin repeat domain	0.6295	2232	1702376
2035	family_706	0.579	ankyrin repeat	0.6298	2331	1716232

Gene_ID	Family	Similarity_score	Product	GC_content	Gene_length_bp	Genomic_position_bp
181	family_737	0.906	transposase domain	0.3103	1563	98552
915	family_737	0.906	transposase	0.3065	1563	1007396
999	family_737	0.906	transposase	0.3131	1584	1112266
1064	family_743	0.548	ankyrin repeat domain	0.6480	1179	1446
1066	family_743	0.548	ankyrin repeat domain	0.6648	1569	7540
1797	family_743	0.548	ankyrin repeat domain	0.6699	1221	1299834
2034	family_743	0.548	ankyrin repeat domain	0.6204	1233	1713922
1065	family_744	0.537	ankyrin repeat domain	0.6926	2163	4706
1617	family_744	0.537	ankyrin repeat domain	0.6517	933	990936
1632	family_744	0.537	ankyrin repeat domain	0.6501	1266	1013066
1783	family_744	0.537	ankyrin repeat domain	0.6725	1536	1272272
1802	family_744	0.537	ankyrin repeat domain	0.6455	1038	1305679
1854	family_744	0.537	ankyrin repeat domain	0.6597	1869	1385658

Gene_ID	Family	Similarity_score	Product	GC_content	Gene_length_bp	Genomic_position_bp
1959	family_744	0.537	ankyrin repeat domain	0.6635	1263	1572709
2368	family_744	0.537	ankyrin repeat domain	0.6573	1278	2274135
1966	family_770	0.504	ankyrin repeat domain	0.6629	1587	1586661
2083	family_770	0.504	ankyrin repeat domain	0.6438	1533	1791397
1909	family_772	0.501	ankyrin repeat domain	0.6648	1635	1478998
2156	family_772	0.501	ankyrin repeat domain	0.6573	1824	1916831
1361	family_977	0.542	hypothetical protein	0.6489	1236	534519
1374	family_977	0.542	hypothetical protein	0.6646	1419	559644

Supplementary Table S4

Table S4. Product-level non-self BLASTp hit rates for all annotated protein categories.

Product	Total_genes	Non_self_hits	Hit_rate_pct	Top_hit_categories
ankyrin repeat domain	130		N/A	
F-box domain	92		N/A	
F-box incomplete domain	26		N/A	
morn repeat domain	44		N/A	

Product	Total_genes	Non_self_hits	Hit_rate_pct	Top_hit_categories
DNA pol3 delta2 superfamily incomplete domain	2		N/A	
Leucin rich repeat domain	1		N/A	
6-phospho-beta-glucosidase incomplete domain	1		N/A	
BTB/POZ domain-containing protein	3		N/A	
Atrophin-1 superfamily incomplete domain	9		N/A	
Phosphohydrolase incomplete domain	1		N/A	
BTB domain	3		N/A	
AICARFT IMPCHas domain	1		N/A	
HPPK domain and Pterin binding domain containing protein	1		N/A	
Dihydroneopterin aldolase	1		N/A	
GTP cyclohydrolase	1		N/A	
glutamine amidotransferase	1		N/A	

Product	Total_genes	Non_self_hits	Hit_rate_pct	Top_hit_categories
Haloacid dehydrogenase	2		N/A	
Collagen	2		N/A	
Complement C1q subcomponent subunit B	3		N/A	
Dihydrofolate reductase	2		N/A	
Von Willebrand factor type A domain	1		N/A	
DNA-3-methyladenine glycosylase	1		N/A	
YjgF/Yer057c/U K114 family protein	2		N/A	
BTB incomplete domain	3		N/A	
NR DBD like superfamily incomplete domain	1		N/A	
morn repeat incomplete domain	6		N/A	
Endopeptidase domain	1		N/A	
Pup ligase superfamily incomplete domain	1		N/A	
SPFH domain containing protein	1		N/A	
Fascin-like incomplete	10		N/A	

Product	Total_genes	Non_self_hits	Hit_rate_pct	Top_hit_categories
domain				
Fascin-like protein	3		N/A	
ZipA superfamily incomplete domain	1		N/A	
Histidine decarboxylase domain	1		N/A	
Ubiquitin	5		N/A	
Esterase FrsA incomplete domain	1		N/A	
Serine/threonine protein kinase	2		N/A	
Kinase VIP2 domain	1		N/A	
DNA pol3 gamma3 superfamily incomplete domain	5		N/A	
Glucose-methanol-choline oxidoreductase	1		N/A	
Herpes BLLF1 superfamily incomplete domain	1		N/A	
YncE incomplete domain	1		N/A	
putative membrane transporter protein	1		N/A	

Product	Total_genes	Non_self_hits	Hit_rate_pct	Top_hit_categories
TF Zn Ribbon incomplete domain	1		N/A	
Retinal superfamily incomplete domain	1		N/A	
Cupin	2		N/A	
AAA ATPase	1		N/A	
SNF2 family DNA dependent ATPase	2		N/A	
LGT superfamily incomplete domain	1		N/A	
PRK00708 superfamily incomplete domain	1		N/A	
ExoD superfamily incomplete domain	1		N/A	

Table truncated; full data (167 rows) available in the repository CSV file.

Supplementary Table S5

Table S5. E-value sensitivity analysis for annotated gene non-self hit rates across four thresholds.

E_value_threshold	Non_self_hits	Hit_rate_pct	Orphan_rate_pct
1e-50	50	9.6	90.4
1e-20	83	15.9	84.1
1e-10	88	16.9	83.1
1e-05	95	18.2	81.8

Supplementary Table S6

Table S6. Complete taxonomy of all 95 non-self BLAST hits with organism, identity, coverage, and genomic position.

Protein_ID	Product	Genomic_position_bp	Strand	Taxonomy_category	Top_hit_organism	Pct_identity	E_value	Query_coverage	In_native_zone
YP_008436418.1	ankyrin repeat domain	1446	+	Giant virus	Cedrat virus kamchatka	54.2	3.25e-112	78.1	No
YP_008436423.1	ankyrin repeat domain	7540	+	Marine algae	Micro monas pusilla CCMP 1545	48.6	2.30e-120	83.3	No
YP_008436426.1	F-box domain	12876	-	Bacteria	Clostridia bacterium	28.5	8.11e-09	41.8	No
YP_008436441.1	ankyrin repeat domain	30576	+	Marine algae	Scytosiphon promiscuus	67.8	1.24e-31	60.4	No
YP_008436573.1	ankyrin repeat domain	148744	+	Giant virus	Acanthamoeba castellanii medusavirus J1	51.8	9.48e-146	95.8	No
YP_008436630.1	BTB incomplete domain	201745	+	Amoebozoa	Acanthamoeba castellanii str.	29.9	2.36e-07	63.7	No

Protein_ID	Product	Genomic_position_bp	Strand	Taxonomy_category	Top_hit_organism	Pct_identity	E_value	Query_coverage	In_native_zone
YP_008437180.2	Acetyltransferase 1 incomplete domain	731926	+	Giant virus	Neff Pacmanvirus S19	49.6	3.04e-39	100.0	No
YP_008437257.1	Intein splicing superfamily domain	793841	-	Amoebozoa	Balamuthia mandrillaris	83.5	2.88e-121	25.6	No
YP_008437323.1	Alpha/beta hydrolase	860261	-	Giant virus	Acanthamoeba castellanii medusavirus J1	60.6	2.61e-76	90.2	No
YP_008437411.1	ankyrin repeat domain	932212	-	Giant virus	Cedrat virus lausannensis	55.6	5.62e-38	59.4	No
YP_008437435.1	ankyrin repeat domain	955237	-	Bacteria	Akkermansia sp.	39.5	1.17e-12	82.6	No
YP_008437437.1	F-box domain	955929	+	Giant virus	Bodotrans virus	49.1	3.22e-45	60.5	No
YP_008437443.1	morn repeat domain	960051	-	Bacteria	Desulfobulbaceae	38.9	5.63e-50	66.5	No

Protein_ID	Product	Genomic_position_bp	Strand	Taxonomy_category	Top_hit_organism	Pct_identity	E_value	Query_coverage	In_native_zone
	n				bacterium				
YP_008437447.1	ankyrin repeat domain	963628	+	Marine algae	Myriotrichia clavaeformis	44.7	1.05e-90	85.7	No
YP_008437448.1	ankyrin repeat domain	965211	+	Marine algae	Micromonas pusilla CCMP 1545	54.7	9.16e-77	70.5	No
YP_008437470.1	ankyrin repeat domain	990936	-	Marine algae	Myriotrichia clavaeformis	55.1	1.32e-75	73.9	No
YP_008437480.1	ankyrin repeat domain	1002303	+	Giant virus	Bodotrans virus	39.1	2.97e-62	67.1	No
YP_008437489.1	ankyrin repeat domain	1013066	-	Marine algae	Ectocarpus fasciculatus	52.8	5.63e-111	76.5	No
YP_008437491.1	ankyrin repeat domain	1014801	+	Marine algae	Dunaliella salina	49.4	7.81e-74	79.1	No
YP_008437493.1	F-box domain	1017150	+	Giant virus	Bodotrans virus	50.9	8.80e-43	54.3	No
YP_0084375	ankyrin repeat	1040791	+	Insect	Acyrtosiphon	59.6	6.78e-44	57.1	No

Protein_ID	Product	Genomic_position_bp	Strand	Taxonomy_category	Top_hit_organism	Pct_identity	E_value	Query_coverage	In_native_zone
21.1	domain				pisum				
YP_008437532.1	ankyrin repeat domain	1049847	+	Other eukaryote	Prymnesium parvum	39.9	3.13e-76	80.0	No
YP_008437543.1	F-box domain	1063923	+	Insect	Acyrtosiphon pisum	34.3	1.28e-35	86.0	No
YP_008437545.1	ankyrin repeat domain	1065526	+	Marine algae	Dunaliella salina	43.5	4.10e-71	51.4	No
YP_008437635.1	Methyltransferase incomplete domain	1151005	+	Giant virus	Mollivirus kamchatka	72.8		97.6	No
YP_008437707.2	serine-threonine kinase	1212260	+	Giant virus	Pithovirus sibericum	47.1		98.1	No
YP_008437767.2	ankyrin repeat domain	1272272	+	Marine algae	Scytosiphon promiscuus	42.5	1.61e-103	92.8	No
YP_008437772.1	morn repeat domain	1280020	+	Bacteria	Desulfohalobacterium	40.6	1.63e-36	43.7	No

Protein_ID	Product	Genomic_position_bp	Strand	Taxonomy_category	Top_hit_organism	Pct_identity	E_value	Query_coverage	In_native_zone
YP_008437782.2	ankyrin repeat domain	1291436	+	Marine algae	Feldmannia mitchelliae	47.2	2.69e-88	69.4	No
YP_008437789.1	ankyrin repeat domain	1299834	+	Giant virus	Cedrat virus kamchatka	51.1	3.16e-105	78.1	No
YP_008437795.1	ankyrin repeat domain	1305679	+	Marine algae	Scytosiphon promiscuus	47.6	2.22e-79	82.3	Yes
YP_008437800.1	ankyrin repeat domain	1312285	+	Giant virus	Cedrat virus duvanney	41.7	1.11e-79	70.2	Yes
YP_008437802.1	ankyrin repeat domain	1318512	-	Other eukaryote	Seminavis robusta	41.7	3.17e-78	84.4	Yes
YP_008437836.1	ankyrin repeat domain	1345828	+	Marine algae	Micromonas pusilla CCMP 1545	42.3	6.70e-17	40.1	Yes
YP_008437837.1	ankyrin repeat domain	1346801	+	Marine algae	Chorisotarpus tenellus	44.1	9.44e-34	88.6	Yes
YP_008437849.2	TLD-containing	1358291	-	Amoebozoa	Acanthamoeba	48.9	2.24e-44	76.2	Yes

Protein_ID	Product	Genomic_position_bp	Strand	Taxonomy_category	Top_hit_organism	Pct_identity	E_value	Query_coverage	In_native_zone
	protein				castellani str. Neff				
YP_008437879.1	ankyrin repeat domain	1385658	+	Marine algae	Micromonas pusilla CCMP 1545	45.5	9.26e-129	92.0	Yes
YP_008437894.1	Ubiquitin	1398635	+	Other virus	Choristoneura diversana nucleopolyhedrovirus	49.5	7.29e-25	35.5	Yes
YP_008437898.2	Nudix hydrolase	1401842	-	Other eukaryote	Helianthus annuus	26.8	6.36e-16	94.6	Yes
YP_008437902.1	ankyrin repeat domain	1405208	-	Amoebozoa	Acanthamoeba castellanii str. Neff	41.0	7.61e-77	95.7	Yes
YP_008437911.1	ankyrin repeat incomplete domain	1413116	-	Insect	Acyrtosiphon pisum	38.1	1.03e-57	69.0	Yes
YP_0084379	ankyrin	1423385	-	Giant virus	Cedrat virus	37.2	2.04e-70	78.6	Yes

Protein_ID	Product	Genomic_position_bp	Strand	Taxonomy_category	Top_hit_organism	Pct_identity	E_value	Query_coverage	In_native_zone
23.1	repeat domain				kamchatka				
YP_008437930.2	Ubiquitin	1431566	+	Amoebozoa	Acanthamoeba castellanii str. Neff	43.7	3.46e-166	99.5	Yes
YP_008438010.1	Alpha/beta hydrolase	1516827	+	Giant virus	Mollivirus shaoxing	84.0		100.0	Yes
YP_008438066.1	ankyrin repeat domain	1572709	-	Marine algae	Desmarestia herbacea	56.7	2.73e-119	78.1	Yes
YP_008438076.1	Capsular polysaccharide synthesis incomplete domain	1580377	-	Giant virus	Mollivirus shaoxing	39.7	4.62e-46	61.9	Yes
YP_008438134.1	ankyrin repeat domain	1635874	-	Bacteria	Cetobacterium sp.	45.7	5.45e-109	61.3	Yes
YP_008438171.1	ankyrin repeat	1675447	-	Amoebozoa	Balamuthia mandri	32.0	2.56e-43	88.2	Yes

Protein_ID	Product	Genomic_position_bp	Strand	Taxonomy_category	Top_hit_organism	Pct_identity	E_value	Query_coverage	In_native_zone
	domain				llaris				
YP_008438181.2	F-box incomplete domain	1687975	-	Marine algae	Ectocarpus siliculosus	37.2	3.45e-10	27.8	Yes
YP_008438183.1	ankyrin repeat domain	1692246	+	Insect	Acyrtosiphon pisum	39.8	1.24e-45	79.6	Yes

Table truncated; full data (95 rows) available in the repository CSV file.

Supplementary Table S7

Table S7. All 9 hypothetical gene non-self hits with taxonomy, identity, and native zone membership.

Protein_ID	Genomic_position_bp	Strand	Taxonomy_category	Top_hit_organism	Pct_identity	E_value	In_native_zone
YP_008438318.2	1838461	-	Amoebozoa	Acanthamoeba castellanii str. Neff	82.4	1.21e-59	Yes
YP_009430099.1	1657937	-	Bacteria	Rariglobus sp.	30.4	2.89e-37	Yes
YP_009430112.1	1822122	+	Amoebozoa	Acanthamoeba castellanii str. Neff	54.0	1.84e-64	Yes
YP_008437859.1	1370653	+	Giant virus	Cedratvirus lena	46.6	6.09e-91	Yes

Protein_ID	Genomic _position _bp	Strand	Taxonom y_catego ry	Top_hit_ organism	Pct_ident ity	E_value	In_native _zone
YP_0084 37883.1	1389102	-	Giant virus	Cedratvir us plubellav i	37.3	1.88e-19	Yes
YP_0084 37808.1	1323333	+	Giant virus	Cedratvir us Zaza IHUMI	46.6	6.16e-26	Yes
YP_0084 38434.1	1966437	+	Giant virus	Molliviru s kamchat ka	41.0	8.87e-79	Yes
YP_0084 37743.1	1250986	+	Bacteria	Powai lake megaviru s	34.0	3.03e-20	No
YP_0084 37273.2	812147	+	Giant virus	Molliviru s kamchat ka	35.5	2.68e-10	No

Supplementary Table S8

Table S8. Cross-genome compositional homogeneity: full GC, GC3, GC12, ENC, and CAI coefficient of variation comparison.

	G		G		G		G		G		G		E		E		C		C	
	C		C		C		C		C		C		N		N		Al		Al	
Ge	_		_		_		_		_		_		_		_		_		_	
n	m		V		G		C		V		G		m		N		V		C	
o	e		_		C		e		3_		_		e		C		_		Al	
m	a		_s		p		_		a		s		p		_s		a		_s	
e	n		d		ct		n		d		ct		n		d		ct		n	
P	0.	0.	6.	1	0.	0.	2	1	0.	0.	1	1	4	1	2	1	0.	0.	3.	1
hi	4	0	0	1	4	0	2.	1	4	0	3.	1	6.	0.	1.	1	7	0	4	1
X	4	2			3	9	6		6	6	8		4	1	8		0	2		
1	7	7			6	8			1	3			5	2			2	3		
7	9	0			8	8			8	7			3	5			6	8		

	G		G		G		G		G		G		E		E		C		C	
	C		C		C		C		C		C		N		N		C		C	
	m		V		m		V		m		V		m		V		m		V	
	C		C		C		C		C		C		C		C		C		C	
	s		p		s		p		s		p		s		p		s		p	
	d		ct		n		ct		n		ct		n		ct		n		ct	
4													1		9					
L	0.	0.	1	7	0.	0.	2	7	0.	0.	1	7	5	5.	1	7	0.	0.	6.	7
a	4	0	4.	3	5	1	0.	3	4	0	3.	3	0.	9	1.	3	7	0	1	3
m	9	7	3		1	0	7		9	6	0		8	3	7		2	4		
b	4	0			0	5			0	3			7	0			6	4		
d	8	7			8	8			4	9			7	8			1	2		
a													6							
M	0.	0.	1	9	0.	0.	2	9	0.	0.	1	9	4	5.	1	9	0.	0.	4.	9
i	2	0	5.	7	2	0	8.	7	3	0	7.	7	1.	4	3.	7	6	0	1	7
m	8	4	1	9	0	5	4	9	3	5	3	9	6	9	2	9	5	2		9
ivi	7	3			2	7			2	7			4	0			7	6		
ru	9	5			4	4			0	4			5	0			4	7		
s													8							
P.	0.	0.	6.	1	0.	0.	9.	1	0.	0.	6.	1	4	5.	1	1	0.	0.	4.	1
s	6	0	1	4	7	0	6	4	6	0	6	4	5.	9	3.	4	7	0	0	4
al	4	3		3	4	7		3	0	3		3	6	1	0	3	8	3		3
in	5	9		0	0	1		0	0	9		0	0	9		0	6	1		0
u	5	3			0	0			6	6			3	1			5	7		
s													8							

Supplementary Table S9

Table S9. Hypothesis testing matrix: evidentiary status of four genome growth models (HGT, gene duplication, mobile element expansion, de novo gene birth) across all 12 modules.

Module	Finding	HGT	Gene_duplication	Mobile_elements	De_novo_birth
2: Genome architecture	GC gap 10.7pp, lowest coding density	Neutral	Neutral	Neutral	Consistent (large intergenic space = substrate)

Module	Finding	HGT	Gene_duplication	Mobile_elements	De_novo_birth
	(67.5%)				
3: Null models	Poly-A/T 15x enriched; AGCT 1500-fold depleted; 88.6% effect-size noise	Neutral	Neutral	Neutral	Consistent (regulatory signals genuine, not artifacts)
4: Codon usage	ORFan GC3 diff +0.015, CAI diff +0.008; upstream cosine 0.9824	Disfavored (no foreign signatures)	Neutral	Neutral	Consistent (compositional homogeneity)
5: Gene prediction	347 Prodigal-only genes; true IG drops to 23.8%	Neutral	Neutral	Neutral	Consistent (unannotated proto-genes exist)
6: Evolutionary classification	GC CV 6.1% (lowest); 3 GMM clusters; no foreign outliers	Disfavored (no compositional heterogeneity)	Neutral	Neutral	Consistent (uniform composition)
7: Gene families	96.8% singletons; 14 domain-repeat families; no tandem clustering; null d=0.06	Neutral	Disfavored (no whole-gene duplication)	Disfavored (no expansion signal)	Consistent (singletons expected)
8: Protein structure	9 structural families; d=1.45 vs null; 41% GC confound; proto-gene	Neutral	Neutral	Neutral	Consistent (structural continuity across tiers)

Module	Finding	HGT	Gene_duplication	Mobile_elements	De_novo_birth
	NN p=0.90				
7b-c: Proto-gene continuum	T1-T5 monotonic gradient; T5 codon p=0.0005; no spatial clustering	Disfavored (no sharp boundaries)	Neutral	Neutral	Supported (gradual maturation)
9: Regulatory signals	+10.3/+9.8p p AT spikes; 82.8% intergenic; T4 p=0.14, T5 p=0.25	Neutral	Neutral	Neutral	Supported (distributed regulatory infrastructure)
10: Transcriptomic validation	74% strand-aware correct (p<10 ⁻⁶); two-pop AT p=0.105	Neutral	Neutral	Neutral	Supported (regulatory signals validated)
11-12: BLASTp homology	81.8% orphan; OR=7.19 (p=7.79e-12); phylogenetic distance gradient	Disfavored (eukaryote-dominated, not multi-kingdom)	Neutral	Neutral	Supported (age-dependent homology gradient)

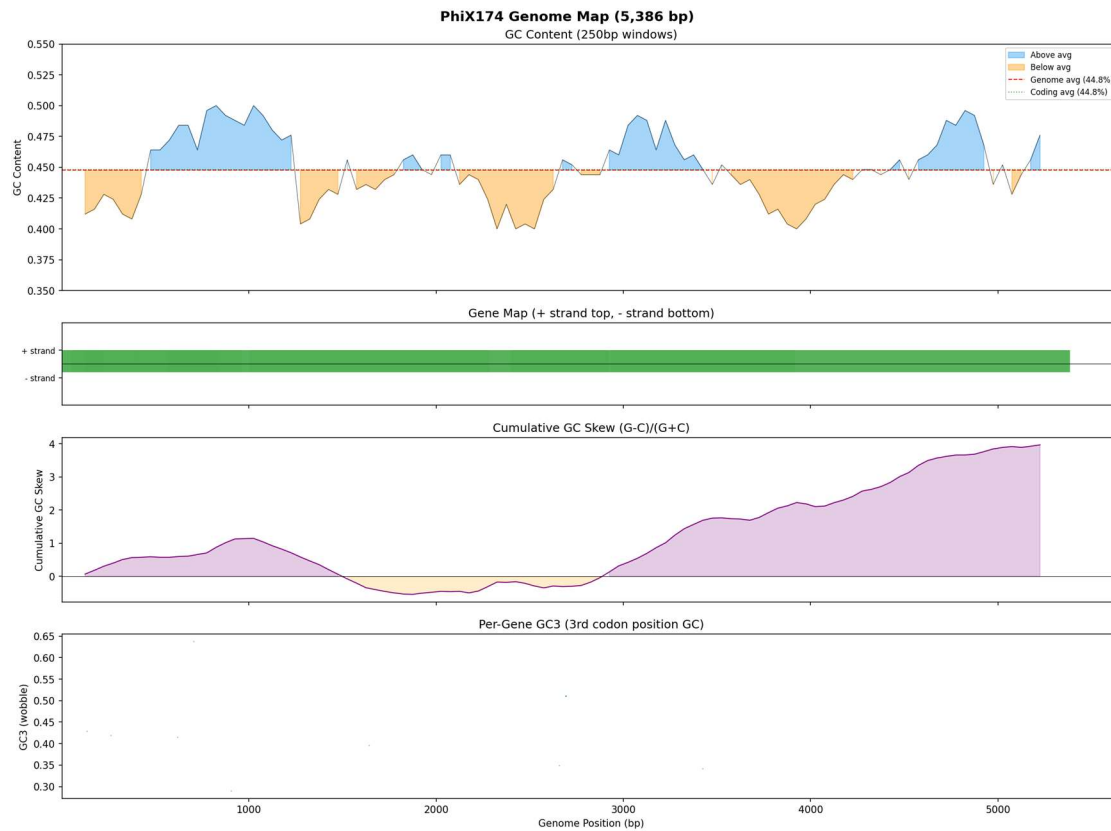
Supplementary Table S10

Table S10. Gene boundary AT signals across genomes: AT content, start and end delta-AT values, and poly-A/T intergenic fraction for Lambda, Mimivirus, and *P. salinus*.

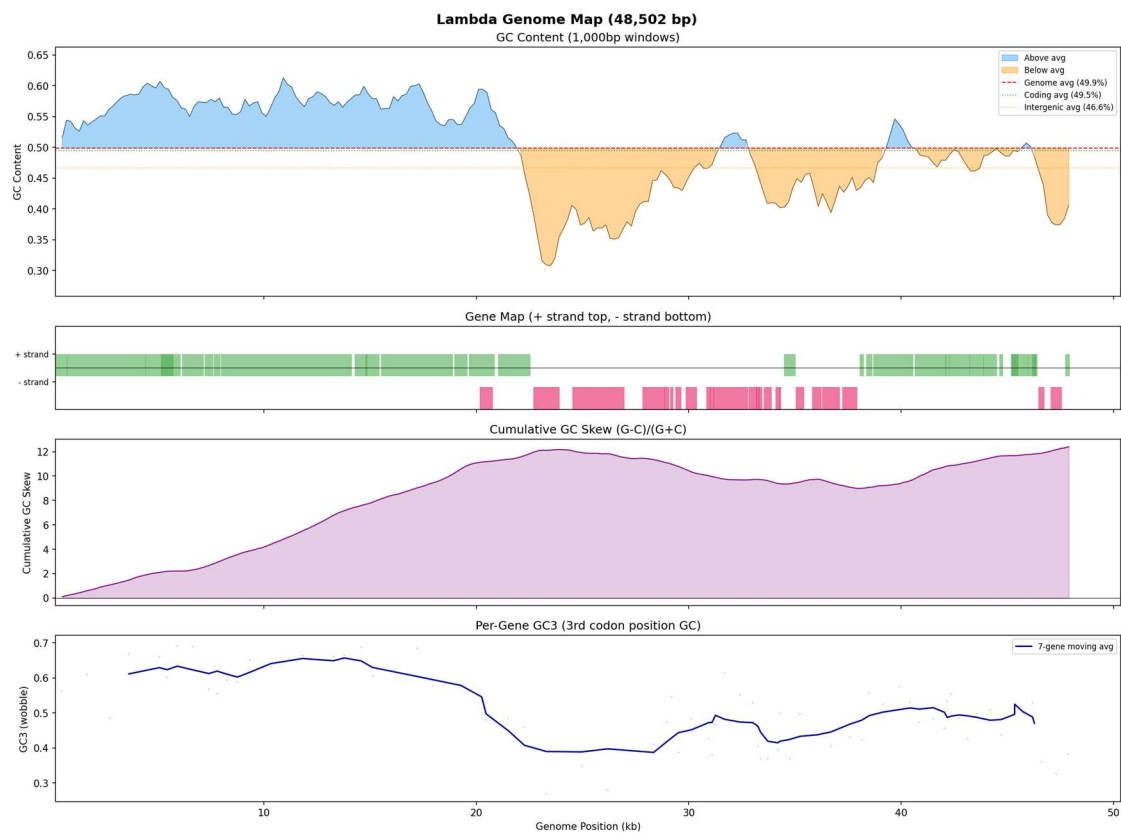
Genome	AT_content	GC_content	Start_delta_AT_pp	End_delta_AT_pp	PolyAT_intergenic_pct
Lambda	50.1%	49.9%	-1.1	+1.6	12-24%
Mimivirus	72.0%	28.0%	+1.1	+2.1	16-17%
<i>P. salinus</i>	38.3%	61.7%	+10.3	+9.8	82.8-83.9%

Supplementary Figure S1

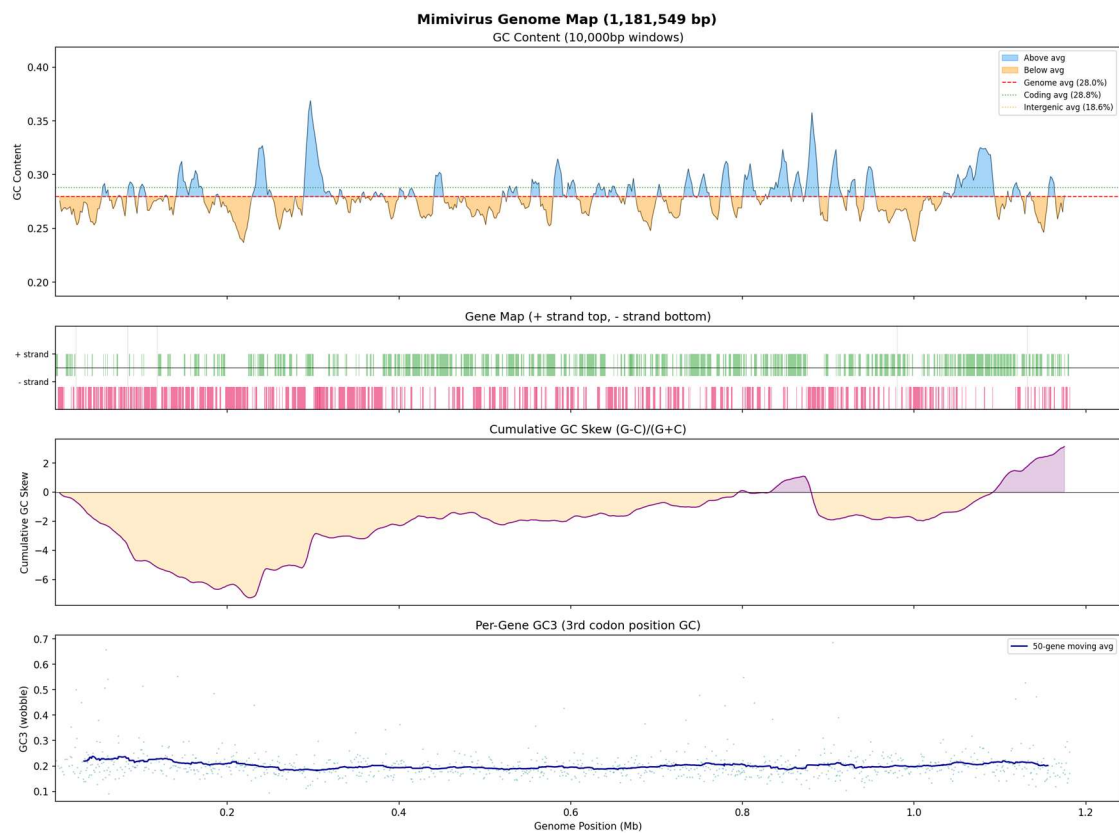
Figure S1. Genome architecture: sliding-window GC profiles, gene density maps, GC skew plots, and strand maps for all four genomes in the comparison ladder.



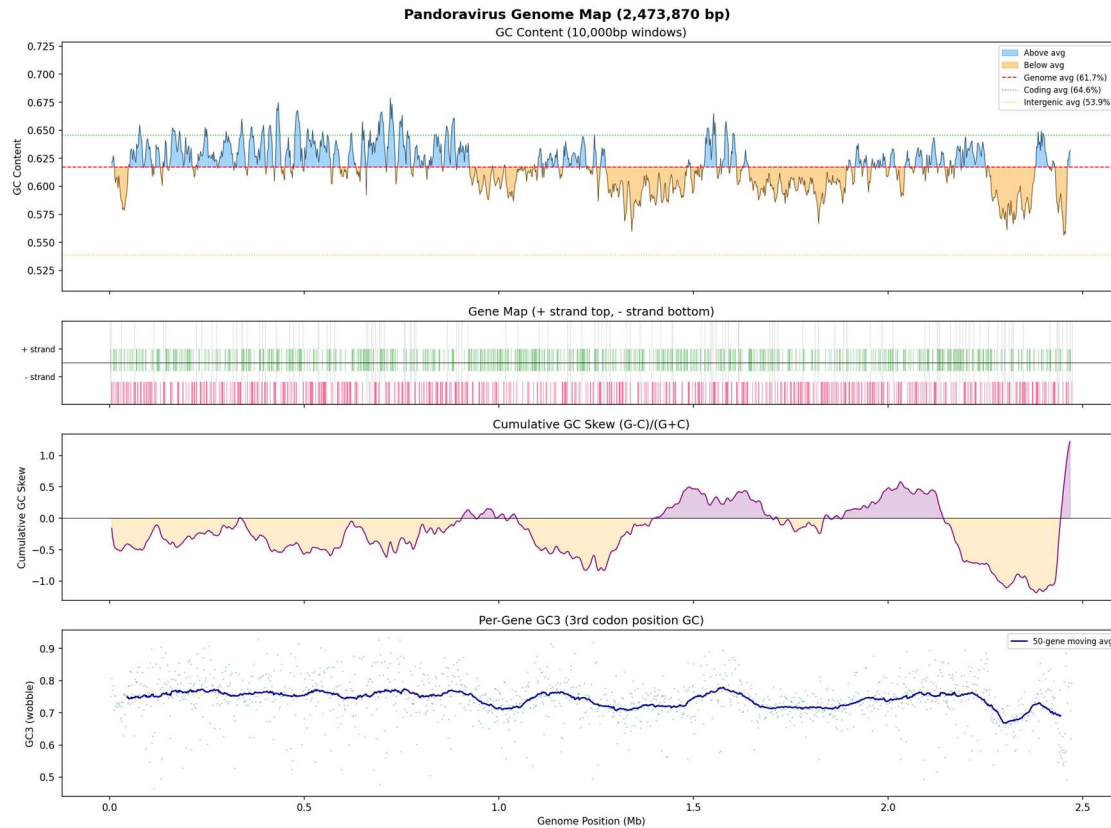
PhiX174 genome map



Lambda genome map



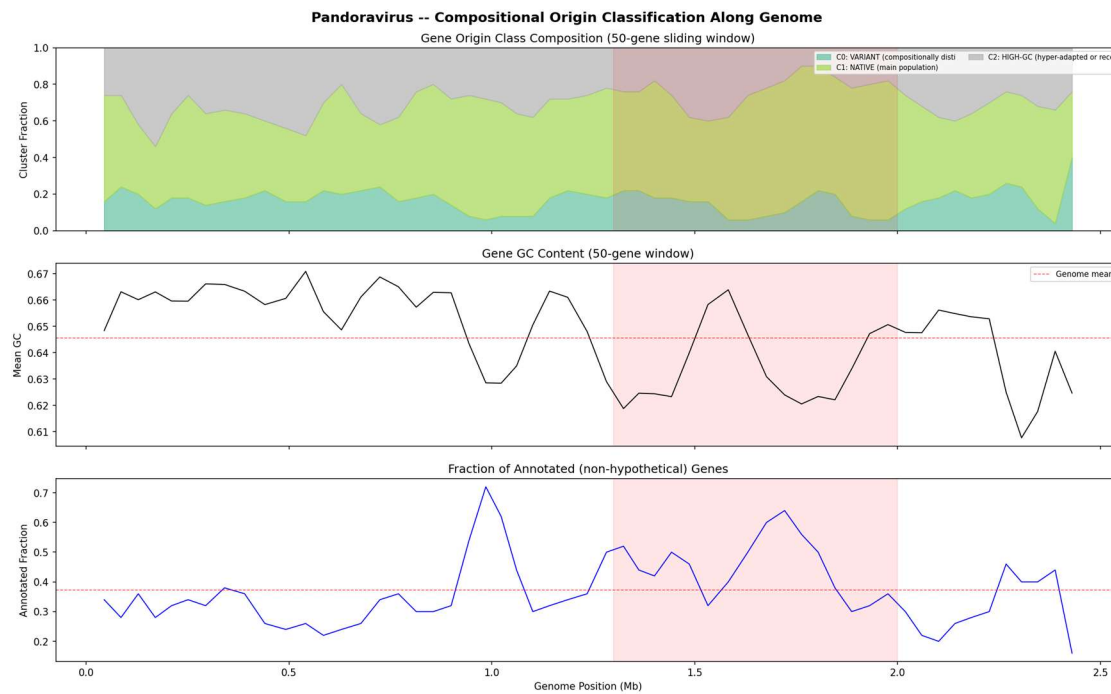
Mimivirus genome map



P. salinus genome map

Supplementary Figure S2

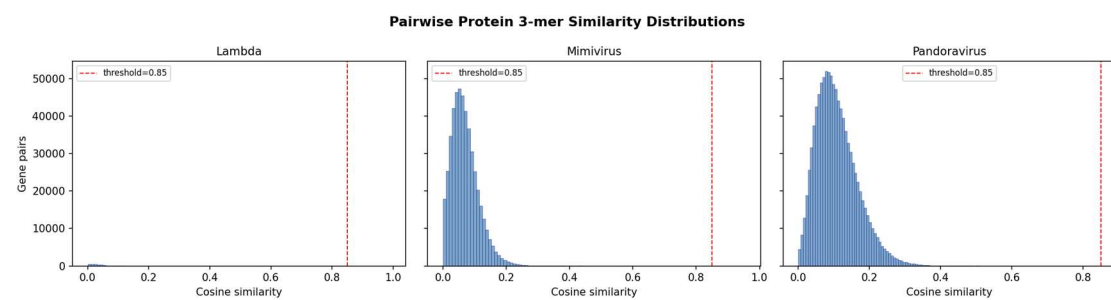
Figure S2. GMM clustering of *P. salinus* genes: three compositional clusters with native zone (1.3-2.0 Mb) overlay, showing cluster composition shift within the zone.



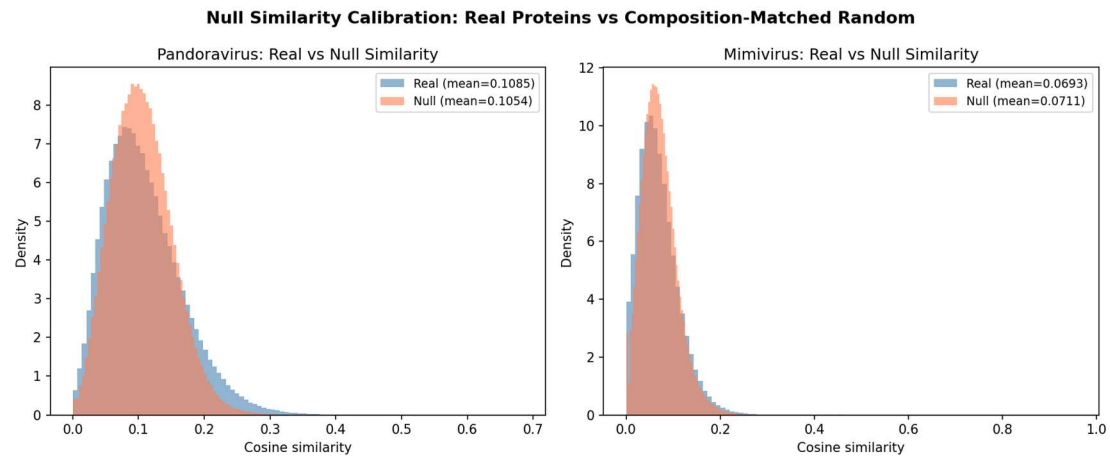
GMM clustering

Supplementary Figure S3

Figure S3. Pairwise protein 3-mer cosine similarity distributions: *P. salinus* vs Mimivirus, with composition-matched null model overlay.



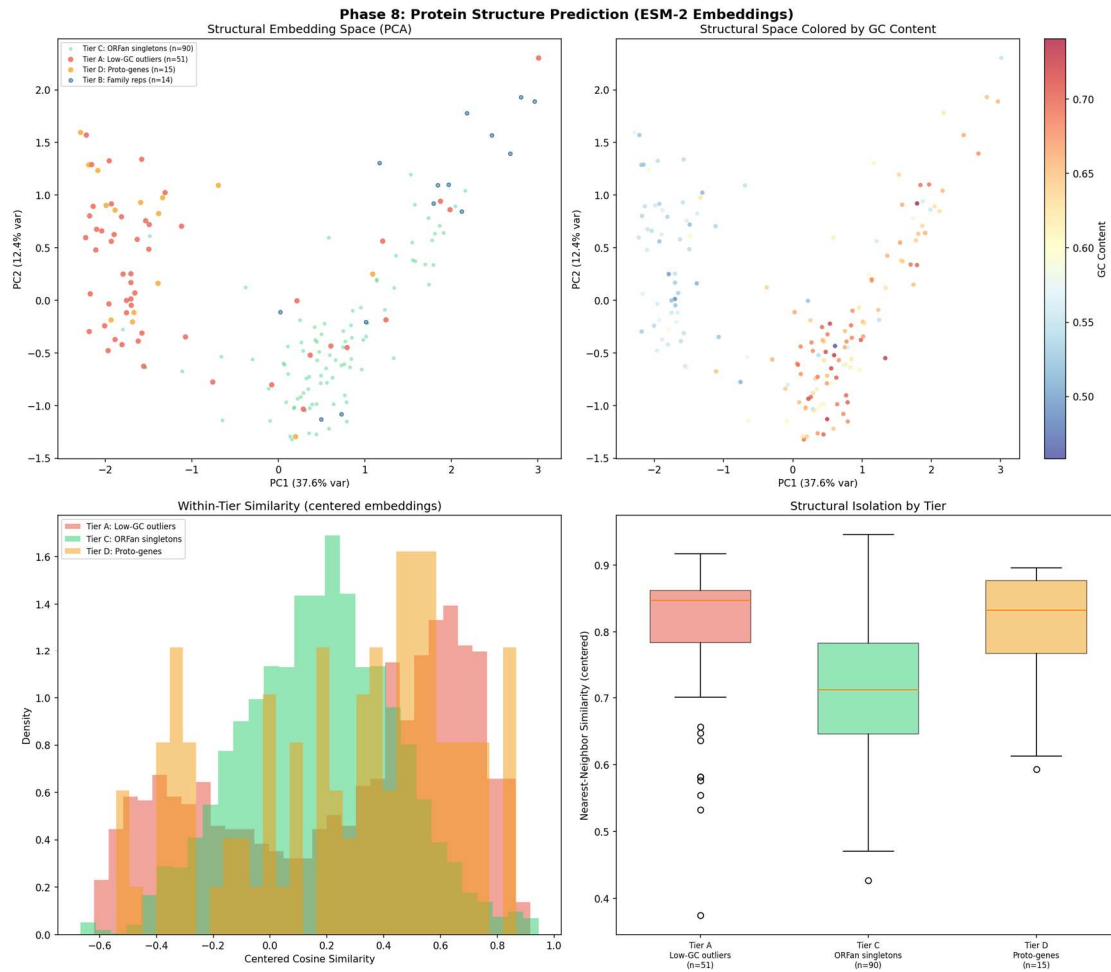
Similarity distributions



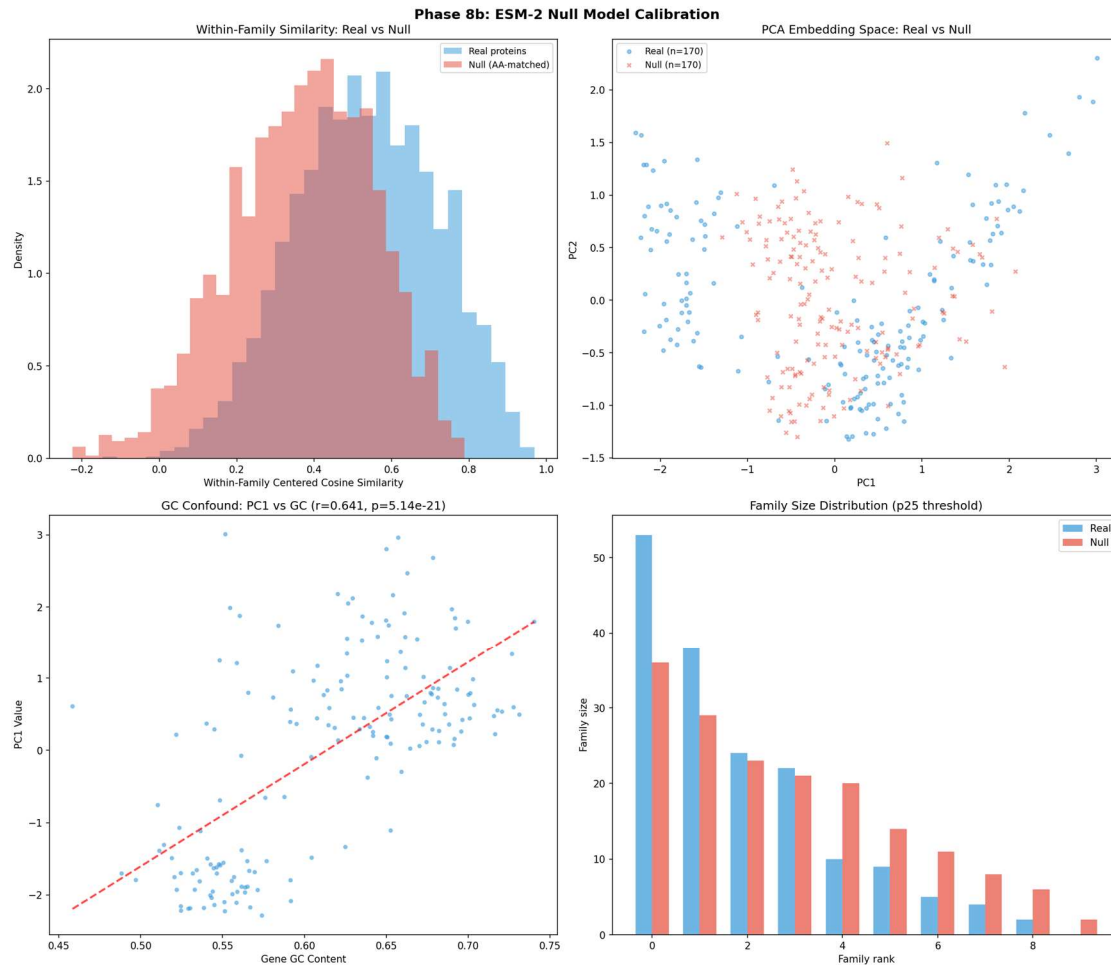
Null calibration

Supplementary Figure S4

Figure S4. ESM-2 PCA embedding space showing structural family assignments, proto-gene positions, GC content overlay, and null calibration.



Structure prediction

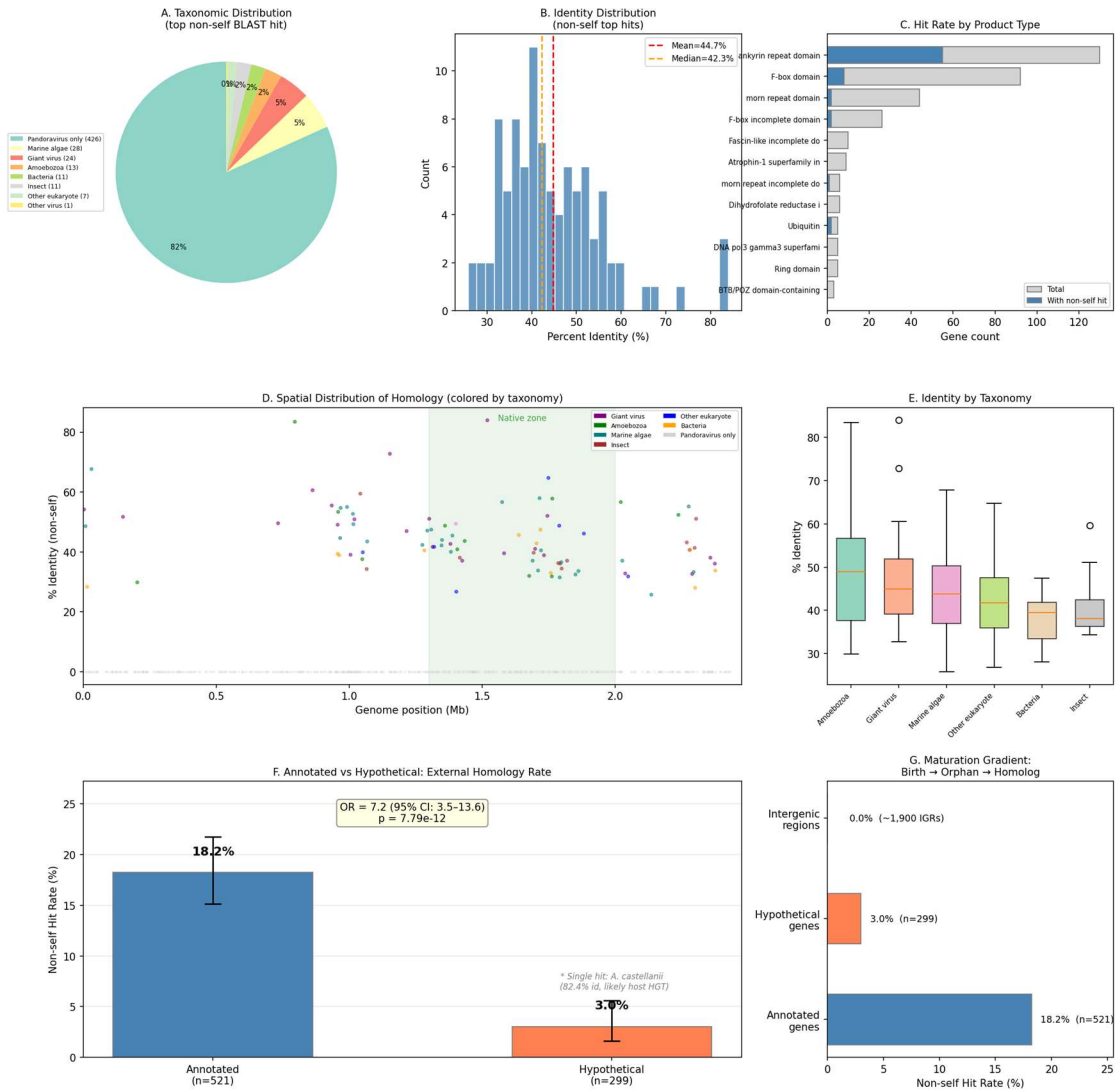


Structure null calibration

Supplementary Figure S5

Figure S5. Full 7-panel Module 11 BLASTp analysis figure.

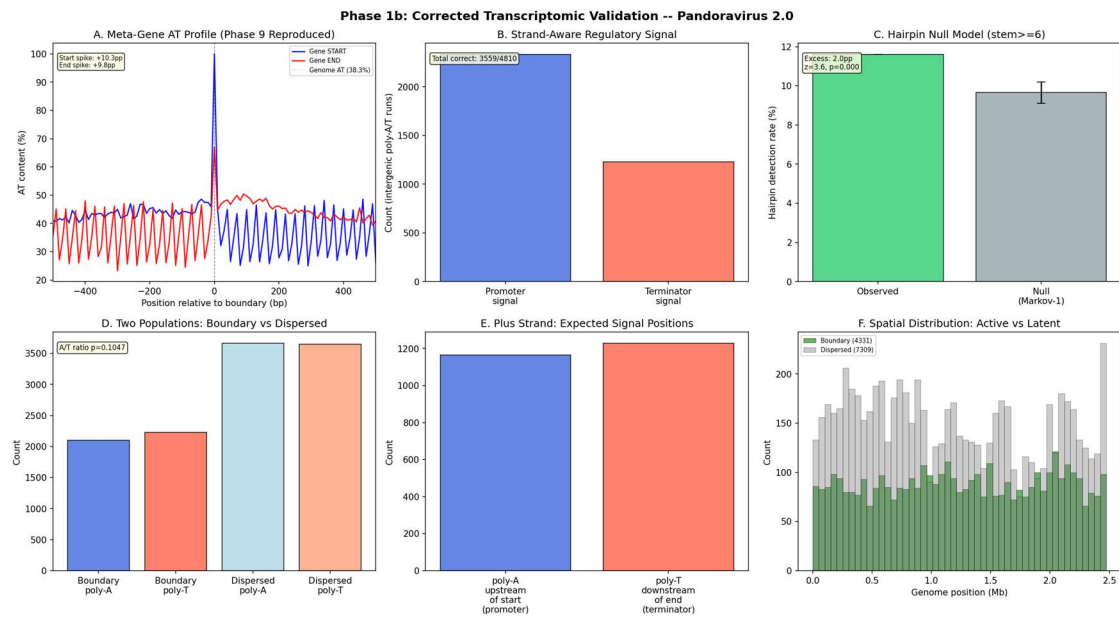
Phase 2: BLASTp Homology Analysis — Pandoravirus salinus



BLASTp analysis

Supplementary Figure S6

Figure S6. Module 10 corrected analysis 6-panel figure.



Transcriptomic validation